ZAVIONIX / ETHOS

Cells
User Manual

A multi-cell battery monitor showing total voltage, individual cell voltages and a voltage-based state-of-charge estimate.

Widget version 3.1.12 | English | 15 October 2026

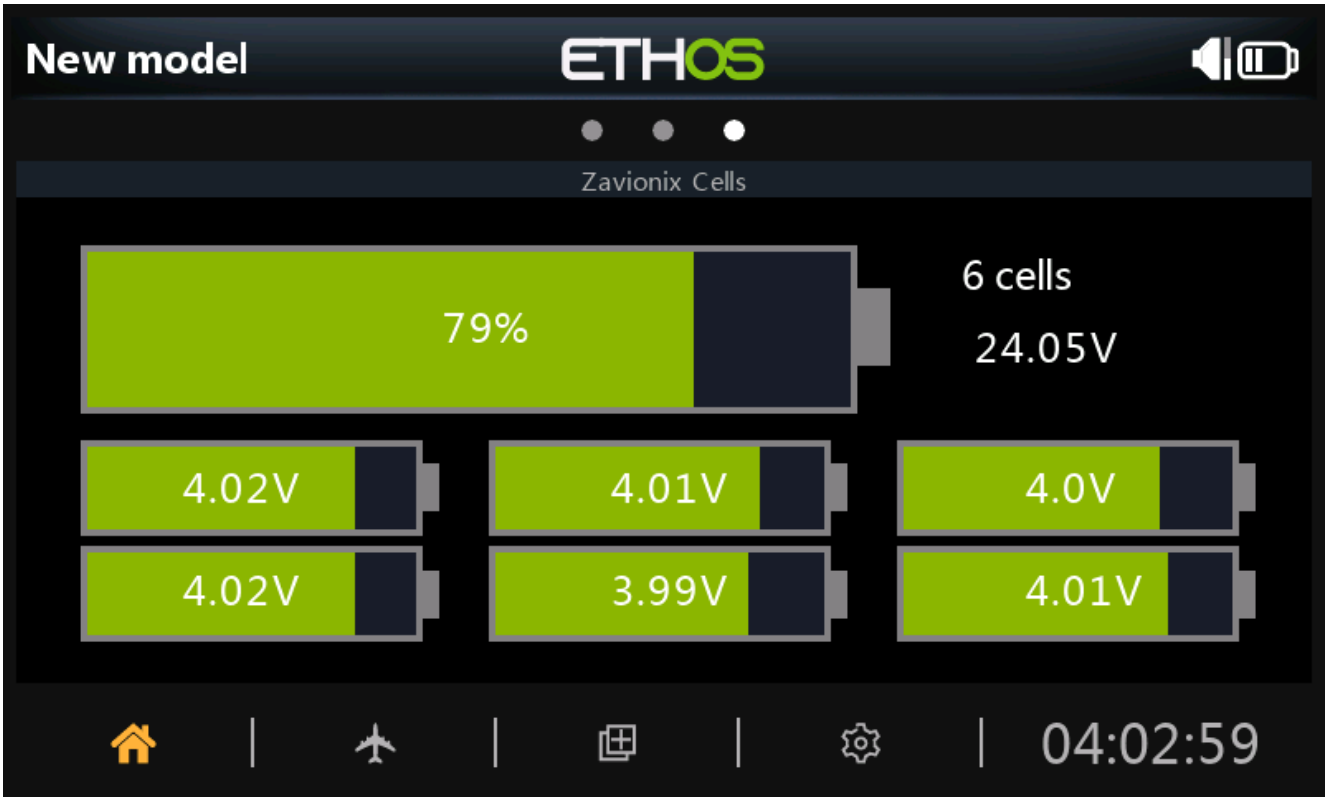


Figure 1. Actual ETHOS simulator rendering with simulated sensor data. The example is not a flight record or a recommended operating setup.

Read this guide in order for first installation. Use the settings and sensor reference when adapting an existing model.

1. Before you begin

A multi-cell battery monitor showing total voltage, individual cell voltages and a voltage-based state-of-charge estimate.

You need a compatible color-screen ETHOS transmitter or simulator, the complete widget folder, and the relevant telemetry sources already available in the model. These captures use the VS Code-cached X20S ETHOS 26.1.3 simulator engine at 800 x 480. Other screens may arrange or scale the content differently.

This manual covers installation, configuration, every captured widget-specific form field, normal use and troubleshooting. Menu labels are written as they appear in this version. Numeric examples are demonstration settings, not model-specific recommendations.

Guide contents

2. Install and add the widget

3. Configure telemetry and read the display

4. Operating guide

5. Settings and sensor reference

6. Troubleshooting and verification

7. Screenshot provenance and version record

2. Install and add the widget

1. Back up the model and existing widget folder. Download the Cells repository ZIP from https://github.com/i3dm/Cells (Code / Download ZIP) or use the appropriate version of the repository.

2. Extract the archive and copy the whole src/Cells folder into the radio storage /scripts directory, producing /scripts/Cells/main.lua. Keep the companion Lua files, images and audio in their supplied relative locations. Do not copy only main.lua.

3. Restart ETHOS or reload Lua through the facilities supported by your firmware. If upgrading, use one complete version of the package rather than mixing old and new module files.

4. Open Configure screens (DISP in the simulator). Add or select a screen and choose a suitable layout; for a full-screen view, long-press its tab and use Change layout. Tap the widget area and select Zavionix Cells from Widget.

5. Open the widget configuration, set the model-specific parameters, and assign telemetry sources. Use the back arrow to return to the dashboard. Reopen the settings and confirm the assignments persist.

Use only the production widget folder on a real transmitter. Documentation sensor fixtures and simulator radio copies are not part of the installation.

3. Configure telemetry and read the display



Figure 2. Widget configuration in ETHOS. Scroll down to reach the remaining settings and source selectors; the reference lists fields below the visible area as well.

Power the intended telemetry hardware in a controlled setup and discover its sensors in the ETHOS model telemetry page. Confirm each value and unit there before assigning it in the widget. Receiver 1/2 and engine left/right inputs must refer to the intended physical source.

A missing value or a No ... label normally means an unassigned or unavailable source. A plausible but incorrect value often means the wrong source, unit or scale. Inspect the source in ETHOS telemetry first; do not compensate for a bad assignment by changing an unrelated capacity or alert setting.

4. Operating guide

Reading individual cells and setting alerts

Select a real cell-voltage sensor such as a compatible FLVSS/FLVS source. The source must expose a cell count and indexed cell values; a generic voltage sensor cannot supply the individual-cell display. This version displays up to eight cells.

The simulator example contains six cells: 4.02, 4.01, 4.00, 4.02, 3.99 and 4.01 V, totalling 24.05 V. The largest-to-smallest spread is 0.03 V. The displayed 79% is a chemistry-dependent voltage estimate, not a measurement of consumed mAh.

Battery Type selects LiPo, HV LiPo, Li-ion (shown as Lion) or LiFe. Set Batt Alert (V) for the total series pack. Set Low Cell Alert (V) for an individual cell and enable Low Cell Alert if required. These thresholds are independent.

The configuration example uses a 10.0 V pack threshold and 3.50 V cell threshold. The pack value is a software example, not a recommended threshold for the pictured six-cell pack. Choose thresholds suitable for the actual chemistry, cell count and operating load.

Use the Font Size setting and widget-area size together. A full-width area makes six or more cells easier to read; reduce the font in a small slot. If a cell count or voltage is absent, inspect the original telemetry source before changing display settings.

5. Settings and sensor reference

The following labels were read from the running widget configuration. Example numbers describe the documentation setup or initialized fields; they are not guaranteed factory defaults for a previously saved model. Source assignments and switches must be chosen for your model. Decimal values below use the units displayed in the form.

Parameters

Menu field	Choices / example / range	Purpose
Battery Type	Lipo; HV Lipo; Lion; LiFe	Select the actual cell chemistry. It selects the voltage-to-percentage lookup, not the charger settings.
Low Battery Alert	On / Off; example On	Enable or disable the named audible alert. Confirm the associated threshold and repeat interval; disabling an alert does not correct bad telemetry.
Batt Alert (V)	Example 10; range 0.1 to 25.2	Total pack low-voltage threshold in volts. This is a pack threshold, not a per-cell value.
Low Cell Alert	On / Off; example Off	Enable or disable the named audible alert. Confirm the associated threshold and repeat interval; disabling an alert does not correct bad telemetry.
Low Cell Alert (V)	Example 3.5; range 2 to 4.2	Individual-cell low-voltage threshold in volts. Enable Low Cell Alert to use it.
Low Battery Alert Rate (s)	Example 5; range 1 to 10	Minimum repeat interval in seconds for the associated alert while its condition persists. The sound files must remain in the widget folder.
Font Size	S; M; L; XL; XXL	Choose S, M, L, XL or XXL to fit the widget area. Reduce the font if individual voltage labels are clipped.

Source assignments

Source selector	Required value and interpretation
Lipo Sensor	A multi-cell telemetry source that supplies both cell count and indexed cell voltages. A scalar pack-voltage source is insufficient.

6. Troubleshooting and verification

Symptom	Checks
Widget absent from selector	Check /scripts/Cells/main.lua and all companion files. Restart ETHOS and inspect Lua messages. Make sure the archive did not create an extra nested folder.
Error after update	Restore a complete matching package version. Check available Lua memory and the exact error in the simulator/radio log; retain the model backup.
Missing / No sensor value	Inspect Model / Telemetry, power the sensor, rediscover if necessary, then reselect the correct source in the widget.

Symptom	Checks
Value has wrong magnitude	Check units, decimals, ECU selection and upstream sensor scaling against a known reference. Do not change warning thresholds to conceal a scaling error.
Values do not change	Compare the original telemetry page. A retained or stale last value is not proof of an active link. Confirm fresh telemetry before use.
Text is clipped	Use a larger widget area/full-screen layout. Cells also has Font Size. Check the intended screen size and version.
No audible alert	Verify enable switch, threshold, repeat interval, telemetry validity, audio files and transmitter volume. Headless screenshots cannot verify the speaker.
Settings appear different	Confirm the selected model, widget instance and, for combined widgets, Widget Display. Each can have its own saved settings.

Model acceptance check

1. Reopen the saved model and verify every configured source and unit.

2. Compare the widget with the underlying telemetry at two or more known values. For twin/battery pairs, vary one side at a time.

3. Check resets, timer controls and any alert conditions relevant to your model in a controlled test. Observe the physical transmitter audio where alerts matter.

4. Verify behavior when a source becomes unavailable and after restoring it. Do not infer source-loss behavior from a static screenshot.

5. Record the final capacities, thresholds, scaling and calibration values with the model backup.

7. Screenshot provenance and version record

The screenshots were produced by the genuine ETHOS X20S 26.1.3 engine (a074fdfd) cached by the VS Code ETHOS extension, driven headlessly through the ethos-tools PR #2 runner. This validates Lua execution and rendering in that engine; it is not a claim that the VS Code webview or every supported transmitter was tested.

Documentation-only copies supplied genuine ETHOS sensor objects and assigned them to the widget. The rendering implementation was retained. Most inputs used DIY sensors refreshed with rawValue; Cells used injected S.Port cell frames to provide an actual multi-cell source. Demonstration settings were initialized independently of the copied model. These fixtures were not installed in the production widget repository.

The simulation method was informed by WingFlight's simulator sensor implementation. The example values are synthetic, not flight measurements. Static captures and configuration inspection do not establish calibrated measurement accuracy, full alert-state coverage, audible playback or physical control performance.

Widget: Cells 3.1.12. Source commit inspected: 0bc454c45e75b976aab70beecd07aef7d20d8c7f. Documentation edition: 15 October 2026. Repository: https://github.com/i3dm/Cells.

Source links

Widget source and updates

ETHOS simulator automation PR

WingFlight simulator sensor approach